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22 August, 2026

Thesis report

Macro-eco-evolutionary perspectives on the Cenozoic history of South American mammals

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PhD thesis submitted to the University of Montpellier

This is an excellent thesis that seeks to explore various aspects of the evolutionary history of South American mammals. It integrates knowledge from paleobiology, macroevolution, macroecology, and paleoclimatology, employing advanced statistical approaches to investigate intriguing questions about the diversification dynamics of South American mammals. The thesis includes an outstanding general introduction, three main chapters, and a comprehensive general discussion. The first and second chapters utilize data from the fossil record to understand the drivers of diversification among different mammal groups. The first chapter investigates the potential existence of a mass extinction event during the Eocene/Oligocene transition and its possible consequences for restructuring mammal faunas in South America. The second chapter focuses on a specific clade of rodents, the superfamily Chinchilloidea, examining the factors that may have contributed to the decline in diversity of this group. The final chapter employs computer simulations to test hypotheses that could explain the asymmetric composition of immigrants observed during the Great American Biotic Interchange.

The **General Introduction** is excellent—well-written and offering a comprehensive review of various topics related to the thesis. It covers the history of paleontology in South America, deep-time biodiversity patterns, factors affecting the completeness of the fossil record and its implications for diversification, various methodological approaches, as well as literature review on diversification dynamics. This reflects the student's engagement and the extensive knowledge gained during their PhD. Given the range of topics discussed in the introduction, it's understandable that some areas aren't explored in depth; however, the narrative conveys that the student has a solid mastery of several subjects, which is quite impressive. I was particularly struck by the general description of South American fauna and the fossil record. The student presented some very nice and informative figures that would be extremely useful for a broad audience of scientists. In fact, the introduction is very well illustrated, featuring a total of 14 figures, most of which are presented clearly and providing valuable information. The student also makes several connections between the thesis



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objectives and the current knowledge reviewed in this section, clearly outlining the goals of the thesis.

In the **first chapter** the student investigates the diversity dynamics of South American Mammals (SAM), to test the hypothesis that the Eocene-Oligocene transition (EOT) also resulted in an important (perhaps mass) extinction event, and faunal turn-over, as previously described for the northern hemisphere. The possibility that extinction regimes were different for tropical lineages vs the non-tropical lineages, as well as the investigation of several potential regulators of biodiversity dynamics (e.g. temperature, diversity dependence, Andean uplift, or diet), were also tested. The chapter represents the first attempt to simultaneously test several hypotheses while explicitly taking in to account the incompleteness of the fossil record. The investigation involved the curation of a large dataset, and the employment of state-to-art methods. The text clearly explains the methods (in the detailed methods section), and the results are presented in nice figures. There is a massive amount of analysis, including several sensitive analyses, that treated the data in different ways. All this states to the fact that doing this chapter clearly served a great learning experience for the student. The results are well presented and quite interesting, suggesting that SAM did not suffer a pulse of high extinction during the EOT, but rather that it suffered a gradual decline. Other interesting results include the suggestion that diversity dependence, temperature and Andean uplift might also be important factors on driving diversification dynamics. Some limitations are discussed (I noticed that a section of limitations that was present in the paper is absent here) but I would suggest the student to further think and explore (in his future work given that this chapter is published) some analysis/data limitations, particularly those regarding the spatial biases of the south American fossil record. The chapter is published in the prestigious journal Proceedings of the National Academy of Science (PNAS) and represents a very nice contribution to the field.

In the **second chapter** the student investigates the diversification dynamics the super family of Caviomorph rodents, the Chinchilloidea. They study the evolution of body size and the potential factors that might have regulated the speciation and extinction regimes of this lineage. The lineage represents a clear example of diversity decline and possesses the largest ever species of rodent, hence it represents a nice study system to study diversification dynamics of declining clades and the evolution of body mass. The chapter has the collaboration of experts in the group and represents an important assemblage of high-quality occurrence and morphological data. It also employs the use of molecular phylogenies to do imputation/augmentation of fossil occurrences for extant species without any fossil record. Looking at all the analytical approaches used in the chapter, I considered that the array of different analysis represented a good opportunity for the student to be trained in different approaches. The discussion is well presented and connects his results to the overall ideas in the field. It also shows the breadth of literature (as well as the introduction) that the



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candidate read to write the chapter. One specific methodological aspect is quite novel (the **imputation/augmentation of virtual fossil occurrences**) but I do have to admit **I am not convinced it is a good idea**. I am looking forward to discussing this during the defense, as it might be that I might have simply misunderstood their approach. I want to emphasize that my disagree with this specific aspect, does not diminish the overall quality of the chapter. This chapter is a very nice contribution to the field. As far as I could tell the chapter has not yet (at least when I received the thesis pdf) been sent for publication, but it is in a nice shape, hence I suspect submission to a journal will be done relatively soon.

In the **third chapter** the student takes a completely different approach and uses spatially explicit eco-evolutionary simulations and paleoclimatic reconstructions to investigate the role of climate on determining the observed asymmetry (more Northern origin lineages in South America, than vice-versa) resulting from the Great American Biotic Interchange (GABI). To achieve this the student used a simulation engine called Gene3sis, developed by Oskar Hagen and collaborators. The main idea was to test the role of paleoclimatic dynamics on driving the GABI asymmetry. Their results suggest that paleoclimate was not a main factor preventing reciprocal faunal exchange. In fact, their results suggest that the intensity of migration from South to North could be even higher than the opposite direction. Hence their results challenge previous ideas and suggest that the tropical areas did not function as a barrier for faunal exchange. The results are well presented, and the discussion engages with the literature. The chapter nicely exposes the limitations of the approach and suggests further aspects that should be investigated (e.g. biotic interactions). The chapter is currently under review in an international journal, and I suspect it will have no trouble on being published.

The thesis ends with a **General Discussion, Conclusions and Perspectives** section. This section nicely summarizes and ties together several of the results and ideas presented in the thesis regarding the evolution of the South American mammal fauna across the Cenozoic. This section also discussed further aspects to be studied, the limitations of the approaches used in the thesis, and the importance of curatorial work when using large datasets to study diversification dynamics. Lastly, it indicates that the student supervised a master's student during this PhD, which the main goals were to reconstruct the most complete phylogeny of the primate lineage Anthropoidea to study the evolutionary history and migration from Africa and Asia to South America. The results suggest that 4 independent events of transoceanic colonization occurred at different moments in time. That served as a very nice experience for the student, and I am sure it helped him mature as a scientist and to learn even more.

In Summary, this is an excellent thesis, with a vast array of approaches and questions. It includes the quantitative assessment of previous hypothesis using cutting



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edge approaches (Chapters 1 and 2), the assembly of impressive new datasets (chapters 1 and 2), and the use of a wide array of different approaches, including those for fitting models to empirical data (Chapters 1 and 2) as well as eco-evolutionary simulation models that are spatially explicit (Chapter 3). It is an excellent contribution to the field of paleobiology, which has greatly contributed to our understanding of the evolution of South American mammals. It has been a pleasure to review the thesis, and I highly recommend Lucas Buffan to defend his PhD thesis.

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